



Your phone app allows Android mirroring

After introducing the ability to beam texts and photos to your PC with Your Phone app, Windows has added Android mirroring feature to it. <https://digit.in/apr19-91>



Chrome gets a native dark mode

Google Chrome officially gets a native dark mode but only for Mac. Windows support is "on the way." <https://digit.in/apr19-92>

available in the market. Then there is the manufacturing. Samsung manufacturers the Exynos processors, only one variant of which is available in the 8nm process (the Exynos 9820), while the others are manufactured using the 10nm process. The latest Snapdragon, A series, Helio and Kirin SoCs are all fabricated using the 7nm process by a single company, called Taiwan Semiconductor Manufacturing Company (TSMC). The first few generations of the A series of chips were fabricated by Samsung, but even the Apple processors are fabricated by TSMC now. In terms of the size, essentially the smaller the size of the chip, the lower the energy footprint.

As Snapdragon provides the SoC for most of the smartphones in the market, the bundled Adreno GPU is the most popular GPU on smartphones. Adreno is an anagram of Radeon, which was the rebranded line for GPUs by Qualcomm after it acquired handheld technology assets from AMD, in 2009, when smartphones were on the verge of becoming mainstream. These GPUs are pretty powerful, and the latest generation can support 8K 360 degree VR playback. The latest generation of Exynos, Helio and Kirin SoCs all use Mali GPUs, which are designed by ARM holdings. The Mali processor architectures are named after places from the Norse mythology, and are in the third generation now, called Bifrost. The previous generations were Midgard and Utgard. The performance and energy efficiency has been improving over the generations. Qualcomm with its Adreno and ARM with Mali together provide GPUs for most of the smartphones in the market today.

There are some exceptions though. Apple uses a four core GPU of its own design. The HTC One X, the Xiaomi Mi3 and the Xolo Play T1000 used NVidia Tegra SoCs, which had ARM CPUs and Maxwell GPUs. The NVidia Tegra SoCs were mostly used in tablets and convertibles. Some older generations of smartphone such as the Samsung Galaxy Nexus, or the

Apple iPhone used PowerVR GPUs, the designs of which were licensed out to various chip manufacturers. About five years ago, the PowerVR licensed SGX GPUs easily outperformed Adreno and Mali GPUs in performance benchmarks.

As you can see, each SoC uses a number of components, many of which are actually provided by the same few companies, and mostly manufactured by two – TSMC and Samsung. As such, their performance depends on the application. Benchmarks provide only a rough idea of the end user experience. It does not make sense to not buy a smartphone just because it uses a certain CPU or a GPU, which may not be as well known or common in the market.



The Qualcomm Snapdragon 855 mobile platform

Each SoC supports a range of communication options, comes with a particular set of image processing capabilities, and can accommodate a certain amount of memory.

DRAM

The three major players in the DRAM market are Samsung, SK Hynix and the Micron Group. While Samsung dominates with a 56% market share, the three of them together provide the RAM to 98.7% to all smartphones. Elpida had somewhere between 14% to 20% share in the market till 2013, but was then acquired by Micron. The three companies have been sued for

allegedly working together to fix the price for the DRAM components.

The original iPhone rolled out with only 128 MB of RAM, which has steadily grown over the years. Just last month, Samsung announced production of 12GB LPDDR4X RAM modules, which are ideal for use with devices that have folding screens or a range of primary camera options. The Samsung Galaxy S10 ships with 12GB of RAM. Now, how much RAM a phone requires is a moot point. In most cases, more is better, especially if you are a multitasker. The applications in their current state of use are stored in the RAM, which is also used for certain operations such as push messaging and fetching emails. Lower RAM means fewer apps will support multi-

tasking, and generally a degradation of performance if too many applications are running. However, there are some companies such as Google and Apple that build the software for the smartphones very well, and generally optimise their operating systems. This means that these phones can provide a good user experience, even while using lesser amounts of RAM in comparison to their competition.

FLASH MEMORY

NAND Flash memory can be considered as the hard drive of the smartphone, and has no moving parts. As such, the capacity has been increasing